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def candidate_elimination(data):
    num_attributes = len(data[0]) - 1

    # Initialize Specific and General boundaries
    S = ['0'] * num_attributes
    G = [['?']] * num_attributes

    # Initialize S with the first positive example
    for row in data:
        if row[-1] == 'Yes':
            S = row[:-1].copy()
            break

    for row in data:
        inputs, label = row[:-1], row[-1]

        if label == 'Yes':
            # Handle positive example
            for i in range(num_attributes):
                if inputs[i] != S[i]:
                    S[i] = '?'

            # Remove from G any hypothesis inconsistent with positive example
            G = [g for g in G if all(g[i] == '?' or g[i] == inputs[i] for i in
range(num_attributes))]

        else:
            # Handle negative example
            G_new = []
            for g in G:
                if not all(g[i] == '?' or g[i] == inputs[i] for i in
range(num_attributes)):
                    G_new.append(g) # Already inconsistent, so keep it
            else:
                # Specialize g
                for i in range(num_attributes):
                    if g[i] == '?' and inputs[i] != S[i]:
                        g_candidate = g.copy()
                        g_candidate[i] = S[i]
                        if g_candidate not in G_new:
                            G_new.append(g_candidate)

            G = G_new

    return S, G

# Example Usage
if __name__ == "__main__":
    dataset = [
        ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
        ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],

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        ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
        ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
    ]

    s_boundary, g_boundary = candidate_elimination(dataset)
    print("Specific Boundary (S):", s_boundary)
    print("General Boundary (G):", g_boundary)

```

## Program 6: Candidate Elimination

Specific Boundary (S): ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']

General Boundary (G): [['Sunny', '?', '?', '?', '?', '?'],  
['?', 'Warm', '?', '?', '?', '?']]

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